

POLYHEXANIDE AND NORWEGIAN SPRUCE RESIN ACIDS FOR SKIN BURNS OF INTERMEDIATE THICKNESS OF THE FACE IN CHILDREN

JAROSŁAW OLSZYNA

Centrum Leczenia Oparzeń, Samodzielny Publiczny Specjalistyczny Zakład Opieki Zdrowotnej „Zdroje”, ul. Mączna 4, 70-780 Szczecin, e-mail: j.olszyna@zoz-zdroje.pl Wpłynęło: 20.08.2019
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ABSTRACT: Introduction Burn injuries are a significant problem for healthcare systems as they lead to permanent damage of the body, its deformation, disability and sometimes may even cause death. Burns occur twice as often in children compared to adults and usually affect small patients up to the age of four. Often, burns not only cause local effects in the form of loss of skin function, but also burn disease. Modern preparations and multi-faceted management of the burn wound have significantly improved wound healing and reduced mortality. Aim The aim of this study was to apply polyhexanide and resinous Norwegian spruce acids in pediatric patients (N=19) who suffered burns of the facial skin of intermediate depth and to determine the effect of these preparations on the wound healing processes, as well as to determine the degree of pain reduction. Material and methods This study uses the case study research method. The following materials were analyzed: medical records of patients treated at the CLO in Szczecin, photographic records, medical interviews with patients and their family, observation of wounds. The FLACC behavioral and VAS visual-analogue scale was used to assess the severity of pain. Conclusions The use of polyhexanide and resinous Norwegian spruce acids resulted in maintaining adequate moisture of the wound and affected the process of granulation and epithelialization. It also reduced pain during hospitalization. These preparations can be used in the TIME treatment strategy. **KEY WORDS:** burns, burns in children, polyhexanide, resinous acids of Norwegian spruce

Introduction

Different types of wounds require various therapeutic approaches. A burn wound is a particular kind of injury where the therapeutic outcome depends on the appropriate treatment applied [9]. Burns in children are among the most severe injuries, occurring most frequently in this patient group between the ages of 3 and 5 years [20]. They account for 50-75% of all burns, with children under 4 years old making up 75% of these cases [4, 7]. In this age group, burns are usually the result of adult negligence, including parents and caregivers (97% of cases). This is due to the excessive activity of children during this period, their curiosity about the world, and their inability to assess danger [17]. Burns remain one of the leading causes of mortality and morbidity in underdeveloped and developing countries, where they are the third most common cause of death in children aged 5 to 14 years [8, 12, 26, 30]. Boys are twice as likely to suffer burns as girls [17]. Most burns are mild or moderate (80%), with the remaining 20% being severe [3, 18]. Recently, there has been significant progress in burn treatment due to an interdisciplinary approach to burn patients. These efforts include rapid wound closure, early fluid resuscitation, and the introduction of nutritional therapy. Preventing acute respiratory distress syndrome and

multiple organ failure, as well as using modern antiseptics, wound healing preparations, and dressings, are also important. In developed countries, such measures have led to a significant decrease in mortality from these injuries [1, 21].

In the studies presented in this paper, a two-component preparation of polyhexanide (PHMB) 0.1% combined with poloxamer was used. PHMB exhibits selective microbicidal action against microorganisms, not human cells responsible for the epidermalization process [14]. It stimulates proliferation, protects keratinocytes, improves microcirculation in wounds, effectively removes biofilm from the wound, prevents its recurrence, and has analgesic and moisturizing effects. It acts against viruses, Gram-positive and Gram-negative bacteria, including *Escherichia coli*, *Staphylococcus aureus* (MRSA – methicillin-resistant *S. aureus*), *Pseudomonas aeruginosa*, VRE (vancomycin-resistant *Enterococcus*), and fungi (*Candida albicans*, *Aspergillus niger*, *Fusarium solani*) [6, 11, 15, 29]. Poloxamer, on the other hand, is a non-ionic surfactant. It removes biofilm from the wound by breaking polysaccharide bonds and bacterial plankton. Its anti-adhesive action is based on reducing surface tension, decreasing the adhesion of microorganisms to the wound surface, thus preventing biofilm formation and infection development. Additionally, poloxamer reduces the intensity of inflammation and the risk of tissue damage [14].

According to Kramer, "the most effective antiseptic is not the strongest one, but the most appropriate one" [14]. The cytological and tissue toxicity of the commonly used combination of octenidine and phenoxyethanol is higher than that of iodophores or polyhexanide and is approximately equal to the toxicity of chlorhexidine [13]. Therefore, PHMB is the first-line antiseptic for critically colonized and infected chronic wounds and burns. It is the only antiseptic substance that additionally stimulates the wound healing process [14]. In cytotoxicity studies by Hirsch et al. on selected antiseptic preparations – Lavasept® (polyhexanide), Prontosan® (polyhexanide), Braunol® (PVP-iodine 7.5%), Octenisept® (octenidine and phenoxyethanol), Betadine® (PVP-iodine 10%) – using the primary human keratinocyte cell line (HaCaT), it was shown that the highest cell survival, and thus the lowest antiseptic toxicity, is associated with PHMB. It was also proven that polyhexanide is the most friendly active substance for dividing cells (fibroblasts and keratinocytes), contributing to wound healing (Fig. 1) [10].

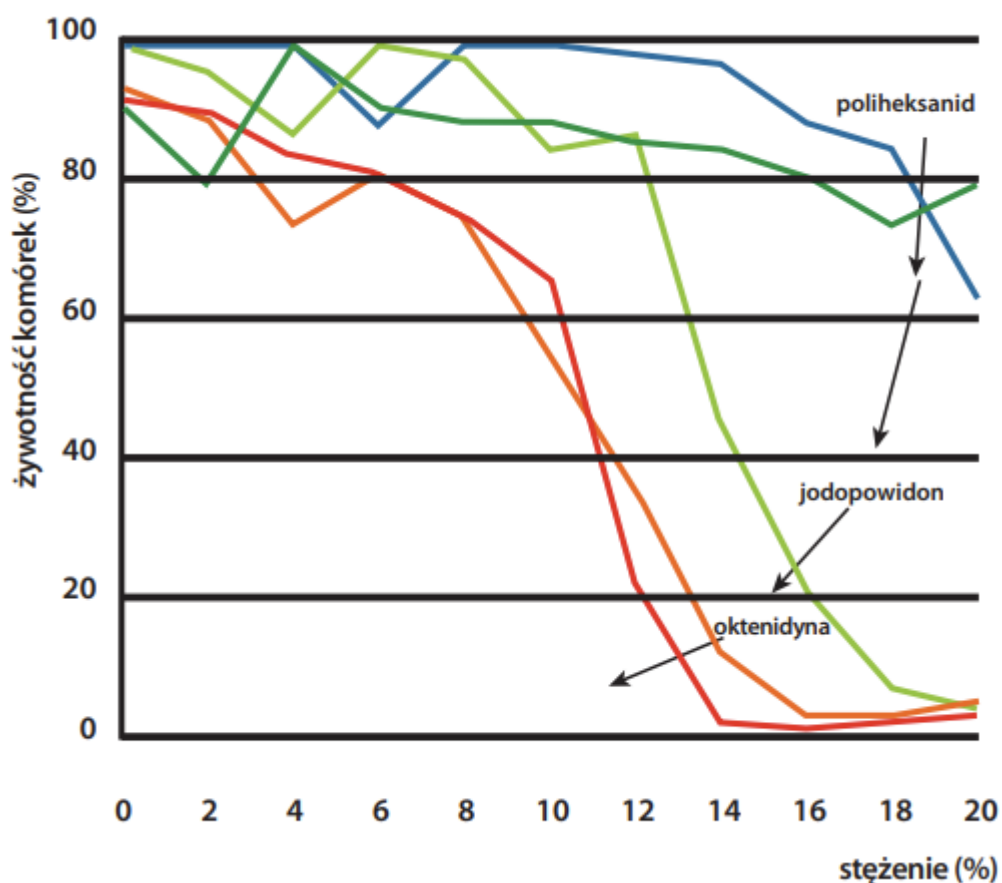


Figure 1. The Impact of Selected Antiseptics on the HaCaT Keratinocyte Cell Line Depending on the Concentration of the Preparation

In therapeutic management, besides PHMB, resin acids from Norwegian spruce (abietic acid, pimaric acid, palustric acid, and lignans belonging to the chemical group of phenolic acids) were also used. These acids exhibit antibacterial activity (also against MRSA, VRE, and fungicidal – including yeast), remove biofilm and prevent its recurrence. They show strong anti-inflammatory and antioxidant properties and support epithelialization. The Norwegian spruce resin acid preparation does not rely on the presence of antibiotics, and therefore does not induce resistance [27].

The aim of this study was to evaluate the use of polyhexanide and Norwegian spruce resin acids in children treated at the Burn Treatment Center in Szczecin, who had facial partial-thickness burns, and to determine the impact of these preparations on wound healing, considering the degree of pain.

Material and Methods

The study involved 19 children hospitalized from May 28, 2018, to June 30, 2019, at the Burn Treatment Center (CLO) in Szczecin, who suffered exclusively from partial-thickness facial burns. All patients sustained their injuries from hot liquids (boiling water, hot tea, hot coffee). The average hospitalization period was 12.2 days (ranging from 5 to 15 days), and the average age of the patients was 1 month (the youngest patient was 1 month old, and the oldest patient was 4.6 years old). The oldest child in the study group was 6 years old, and the youngest was 5 months old. Most of the study group were boys (n=13) (68% boys vs. 32% girls). The characteristics of the study group are presented in Table 1.

Table 1. Demographic and Clinical Characteristics of the Study Group (N=19)

Variable	Value (Percentage)
Age (years, months)	1 year 11/12 months $\pm$ 8/12 months
Gender (N)	
Female	6 (32%)
Male	13 (68%)
Average hospitalization time (days)	11.2 $\pm$ 1.7
Average duration of PHMB use (days)	10.1 $\pm$ 1.1
Average duration of Norwegian spruce resin acids use (days)	10.3 $\pm$ 0.9
Etiology of burn	
Boiling water	5 (26%)
Hot tea	8 (42%)
Hot coffee	6 (32%)

Quantitative variables are presented as mean $\pm$ standard deviation; N – total number of subjects; n – number of units in subgroups.

PHMB gel preparations were applied to the facial burns of the patients several times a day to moisturize the wound, and polyhexanide in liquid form was used once daily to clean the wounds. Additionally, a preparation containing Norwegian spruce resin acids was applied 2–3 times daily.

To assess the intensity of pain, the FLACC (Face, Legs, Activity, Cry, Consolability) behavioral scale was used. This scale evaluates five parameters: facial expression, leg position, activity, crying, and consolability. The values obtained range from 0 to 10 points. This scale does not assess vital parameters, and if a value of ≥ 3 is obtained, pain

management treatment is required (Table 2) [22, 24]. For two subjects (5 and 6 years old), the Visual Analog Scale (VAS) was used. In the VAS, pain is assessed on a 10-point scale, where 0 represents no pain and 10 represents the most intense pain imaginable.

Table 2. FLACC Behavioral Pain Assessment Scale

Category	Score	Assessment
Facial Expression (Face)	0	No particular expression or smiling
	1	Occasional grimace, frown, sad, disinterested
	2	Frequent or constant grimace, quivering chin
Leg Position (Legs)	0	Relaxed, normal position
	1	Uneasy, tense
	2	Kicking, drawn up
Activity (Activity)	0	Lying quietly, normal position, moves easily
	1	Squirming, shifting back and forth, tense
	2	Arched, rigid, or jerking
Crying (Cry)	0	No cry
	1	Moans or whimpers, occasional complaint
	2	Crying steadily, screams or sobs, frequent complaints
Consolability (Consolability)	0	Content, relaxed
	1	Reassured by occasional touching, hugging, or talking to; distractible
	2	Difficult to console or comfort

All patients participating in the study received regular pain management treatment (paracetamol). If the FLACC or VAS score was ≥ 3 , additional pain medication was administered. Pain intensity in patients was assessed on days 1, 2, 4, 6, and 10, and on the day of discharge. The scales were used on days when the children did not undergo dressing changes involving analgesic-sedative drugs (wound cleaning under general anesthesia was performed on day 3).

Results

During the use of PHMB and Norwegian spruce resin acids preparations, the greatest reduction in pain among patients was observed on the 4th day (FLACC and VAS – 2.3). The patients received additional pain medications (most commonly ibuprofen), primarily during the first 2 days of hospitalization. On the 4th day of hospitalization, only four young patients (26%) required pain-reducing medications. By the 6th day, eleven patients (73%) had a FLACC and VAS score of zero (Table 3).

Table 3. Assessment of Pain Intensity Reduction Using the FLACC and VAS Scales on Specific Hospitalization Days

Variable	Day 1	Day 2	Day 4	Day 6	Day 10	Discharge Day
Total number of patients (N= 19)	19	19	14	15	18	19
Average pain intensity (FLACC and VAS Scales)	3.9	3.7	2.3	1.2	0.6	0.5
Standard deviation (SD)	0.45	0.47	0.44	0.30	0.11	0.10

While using PHMB combined with Norwegian spruce resin acids, good hydration of the burn wound was observed. The preparations used in the treatment contributed to the proper course of secondary healing (granulation), also known as *sanatio per secundam intentionem vel granulationem*, and epithelialization. These products align with the foundation of the modern, widely accepted concept of proper wound bed preparation [25]. They provide optimal environmental conditions in the wound bed, generating a proper course of keratinocyte proliferation (Figure 2).



Figure 2. Effect of Using PHMB and Norwegian Spruce Resin Acids for 12 Days in a 6-Year-Old Boy

PHMB and Norwegian spruce resin acids preparations adhere to the principles of the TIME strategy. The goal of wound treatment is to achieve control over the wound environment. Before applying therapeutic products, it is necessary to prepare the base by cleaning the wound. PHMB and Norwegian spruce resin acids products cleanse the wound, protect it from infection, and prepare the wound bed, ensuring optimal conditions for granulation and epithelialization.

In recent years, the number of children treated at the Burn Treatment Center has increased. There has also been an increase in the percentage of patients with facial burns (Table 4). Over 40% of patients treated at the CLO suffered facial burns. In cases of injuries to this part of the body, dressings are relatively rarely used (due to the inability to stabilize the dressing), so it seems that the appropriate form of preparation is an ointment. Therefore, the use of PHMB and Norwegian spruce resin acids preparations is justified in the treatment of facial burns in pediatric patients.

Table 4. Statistics of the Number of CLO Patients and the Percentage of Patients with Facial Burns or Facial Burns with Other Body Parts

Year	Number of Hospitalized Patients	Percentage of Patients with Facial Burns or Facial Burns with Other Body Parts
2017	68	28%*
2018	92	39%*
Up to 30.06.2019	56	43%*

*Percentage of patients with facial burns or facial burns along with other body parts

Discussion and Conclusions

In multicenter randomized clinical trials conducted from 2014 to 2016, it was demonstrated that polyhexamethylene biguanide (PHMB) accelerates re-epithelialization and epidermalization. Within two days, it reduced erythema and eliminated infections in second-degree burns, with an average treatment time of 11.2 days. PHMB is recommended as the first-choice antiseptic for treating colonized wounds and burns and for preventing surgical site infections [28, 30]. It has a beneficial effect on granulation tissue formation in wounds, which accelerates the healing process and leads to significant or complete pain reduction in patients [5, 23]. The improvement in hospitalization comfort associated with the alleviation of pain is particularly important in young patients. No existing publications have shown microbial resistance to PHMB preparations [16, 28]. For centuries, spruce resin has been used as a balm in northern Sweden and Finland. It has

been documented that this ointment has bacteriostatic properties (likely related to the solubilization of the bacterial cell wall) and is effective in treating bedsores, burns, and wound infections [2].

In this study, the use of PHMB and Norwegian spruce resin acids in children contributed to maintaining appropriate wound moisture, influencing the processes of granulation and epithelialization, and reducing pain during hospitalization.

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